

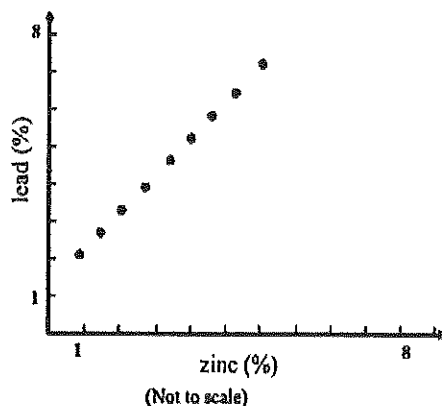
GATE SOLUTION IN MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

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- Q.1** A scatter plot prepared using a set of values of lead and zinc from a lead-zinc deposit is shown in figure below. The value of correlation coefficient is



- (A) 1.0 (B) 0.7 (C) 0.5 (D) 0

Answer (A)

Solution .

By looking at the scatter points, it can be seen that all of the points are very close to being along a straight line . So there is a perfect positive correlation between lead and zinc from lead – zinc deposit.

Therefore,

$$\text{Correlation coefficient } (r) = + 1$$

- Q.2** The value of $\lim_{x \rightarrow 0} \frac{1}{x} \{ \sqrt{1+x} - \sqrt{1-x} \}$ is

- (A) 0 (B) 1 (C) 2 (D) 3

Answer (B)

Solution .

$$f(x) = \lim_{x \rightarrow 0} \frac{1}{x} \{ \sqrt{1+x} - \sqrt{1-x} \}$$

$$f(x) = \lim_{x \rightarrow 0} \frac{1}{x} \left\{ \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \times \sqrt{1+x} + \sqrt{1-x} \right\}$$

$$f(x) = \lim_{x \rightarrow 0} \frac{1}{x} \left\{ \frac{(\sqrt{1+x})^2 - (\sqrt{1-x})^2}{\sqrt{1+x} + \sqrt{1-x}} \right\}$$

$$\therefore f(x) = \lim_{x \rightarrow 0} \frac{1}{x} \left\{ \frac{2x}{\sqrt{1+x} + \sqrt{1-x}} \right\}$$

$$\therefore f(x) = \lim_{x \rightarrow 0} \left\{ \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \right\}$$

$$\therefore f(x) = \frac{2}{2} = 1$$

Q.3 The infinite series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$, is

(A) convergent (B) divergent (C) oscillatory (D) semi-convergent

Answer (A)

Solution .

Given infinite series , $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \dots \dots \infty$

$$\text{Or } \sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \dots \dots \infty$$

$$\text{So, } r = \sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{2^\infty} = 0$$

$\therefore r < 1$, the series is convergent.

Q.4 The largest area of a rectangular shaft for a given constant perimeter is obtained when length is

(A) 2.5 times of breadth (B) 1.5 times of breadth
(C) 2 times of breadth (D) equal to breadth

Answer (D)

Solution .

Perimeter for a rectangular shaft is given by:-

$$\text{Perimeter} = 2(\text{length} + \text{breadth}) \dots \dots \dots (1)$$

Let take perimeter is x

From equation (1)

$$\text{length} + \text{breadth} = 0.5 x \dots \dots \dots (2)$$

Case -1. When length is 2.5 times of breadth

from equa. (2)

$$2.5 \text{ breadth} + \text{breadth} = 0.5 x$$

$$\therefore \text{breadth} = 0.14 x$$

$$\therefore \text{length} = 0.36 x$$

$$\therefore \text{Area} = 0.0504 x^2$$

Case -2. When length is 1.5 times of breadth

from equa. (2)

$$1.5 \text{ breadth} + \text{breadth} = 0.5 x$$

$$\therefore \text{breadth} = 0.2 x$$

$$\therefore \text{length} = 0.3 x$$

$$\therefore \text{Area} = 0.06 x^2$$

Case -3. When length is 2 times of breadth

from equa. (2)

$$2 \text{ breadth} + \text{breadth} = 0.5 x$$

$$\therefore \text{breadth} = 0.17 x$$

$$\therefore \text{length} = 0.34 x$$

$$\therefore \text{Area} = 0.0578 x^2$$

Case -4. When length is equal to the breadth

from equa. (2)

$$\text{breadth} + \text{breadth} = 0.5 x$$

$$\therefore \text{breadth} = 0.25 x$$

$$\therefore \text{length} = 0.25 x$$

$$\therefore \text{Area} = 0.063 x^2$$

Hence , In case of 4, Area of rectangular will be largest.

Q.5 A drive shaft of an engine develops torque of 500 N-m. It rotates at a constant speed of 50 rpm. The power transmitted by the shaft in kW is

- (A) 1.46 (B) 2.05 (C) 2.62 (D) 4.32

Answer (C)

Solution.

Relation of Power (P), Torque (T) and Rotational Speed (ω) are given by-

$$P = \omega . T$$

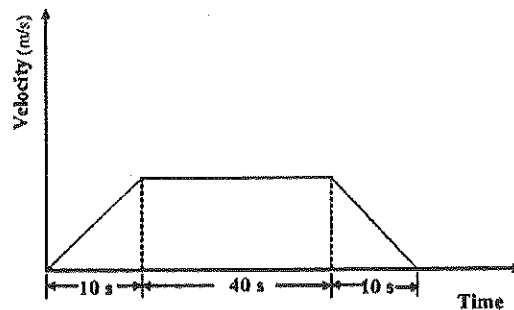
$$\text{Or } P = \frac{2\pi N}{60} . T$$

$$\therefore P = \frac{2\pi N}{60} . T = \frac{2 \times 3.14 \times 50}{60} \times 500$$

$$\therefore P = 2616.7 \text{ watt}$$

$$\therefore P = 2.62 \text{ kW}$$

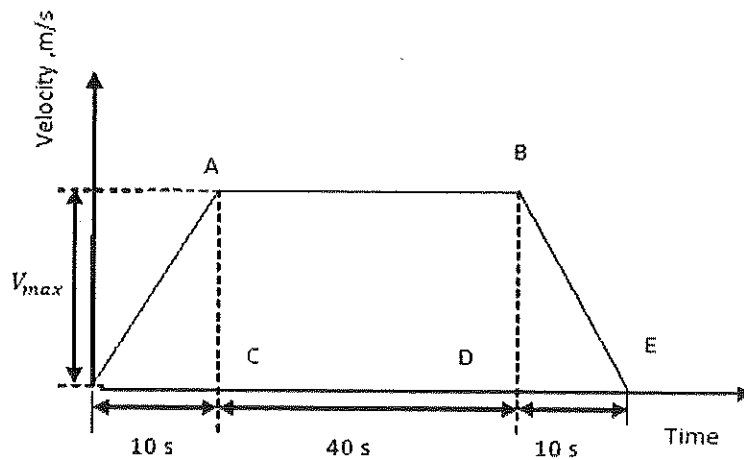
Q.6 A mine winder cage traveling 450 m from pit bottom to pit top is following a three period duty cycle as shown in the figure below. The maximum velocity attained by the cage in m/s is



- (A) 7.5 (B) 9.0 (C) 11.0 (D) 12.0

Answer (B)

Solution .



Here, at point B, the cage will attain maximum velocity. Hence, maximum velocity attained by the cage is-

$$V_{max} = \frac{450}{50} = 9 \text{ m/sec}$$

Q.7 Stress concentration at a point on the wall of a vertical shaft results in a compressive stress of 59.66 MPa. The wall rock mass has an unconfined compressive strength of 89.49 MPa. The safety factor of the shaft wall at the point is

- (A) 0.67 (B) 0.86 (C) 1.23 (D) 1.50

Answer (D)

Solution .

$$\text{Safety factor} = \frac{\text{Strength}}{\text{stress}}$$

$$\text{Safety factor} = \frac{89.49}{59.66}$$

$$\text{Safety factor} = 1.5$$

Q.8 A core sample of 54 mm diameter having Young's modulus of 68.97 GPa fails in uniaxial compression at 0.1% axial strain. The axial load at failure in kN is

- (A) 158.00 (B) 68.97 (C) 58.00 (D) 15.80

Answer (A)

Solution .

We have,

$$\text{Stress} = \text{young modulus} \times \text{strain}$$

$$\text{Or } \frac{\text{Axial load}}{\text{Area}} = \text{young modulus} \times \text{axial strain}$$

$$\therefore \frac{\text{Axial load}}{3.14 \times (27 \times 10^{-3})^2} = 68.97 \times 10^9 \times 1 \times 10^{-3} [\because \text{Axial strain} = 0.1\% = 1 \times 10^{-3}]$$

$$\therefore \text{Axial load} = 68.97 \times 10^9 \times 1 \times 10^{-3} \times 3.14 \times (27 \times 10^{-3})^2 \text{ N}$$

$$\therefore \text{Axial load} = 157876 \text{ N}$$

$$\therefore \text{Axial load} = 158 \text{ kN}$$

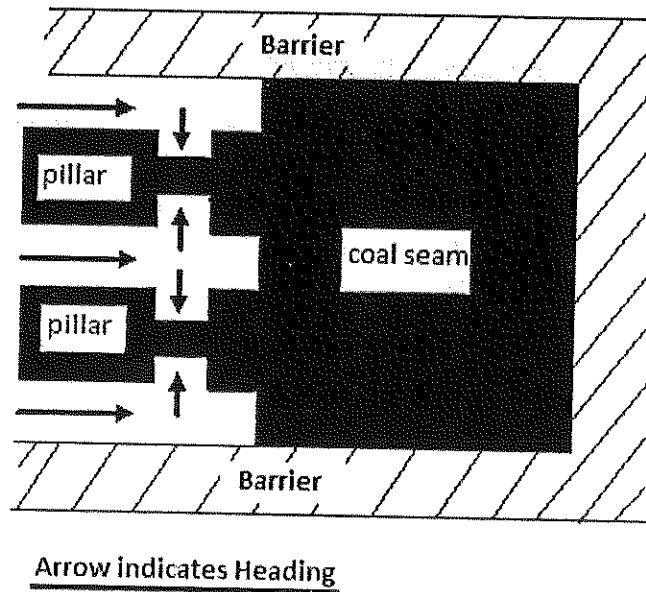
Q.9 The maximum number of coal faces in an underground bord and pillar development district is

13. The number of headings in the district is

- (A) 3 (B) 5 (C) 6 (D) 7

Answer (B)

Solution .



From above figure, relation between number of face (m) and number of heading (n) are given by-

$$m = 3 \times n - 2$$

$$\therefore 13 = 3 \times n - 2$$

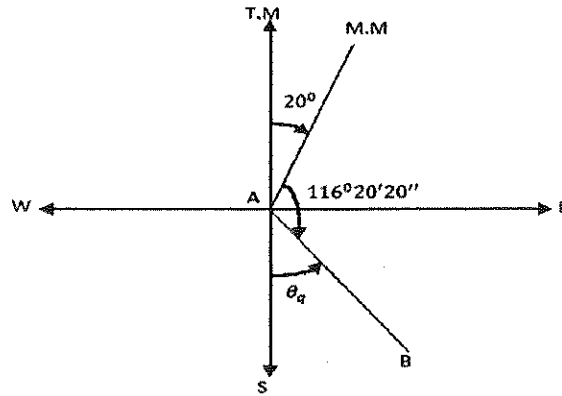
$$\therefore n = 5$$

Q.10 The whole circle bearing of the line AB is $116^\circ 20' 20''$. If there exists an east declination of 20° , the true quadrantal bearing of line AB is

- | | |
|--------------------------|--------------------------|
| (A) $S41^\circ 59'40''E$ | (B) $S43^\circ 39'40''E$ |
| (C) $S45^\circ 59'40''W$ | (D) $S47^\circ 59'40''W$ |

Answer (B)

Solution .



T.M. = True Meridian

M.M. = Magnetic Meridian

$$\text{True bearing of line AB} = 116^{\circ}20'20'' + 20^{\circ}0'0'' = 136^{\circ}20'20''$$

$$\therefore \text{ True quadrantal bearing of line AB} = S (180^{\circ}0'0'' - 136^{\circ}20'20'') E$$

$$\therefore \text{ True quadrantal bearing of line AB} = S 43^{\circ}39'40'' E$$

Q.11 It is proposed to connect two straights of a road by a simple circular curve. If the maximum speed of the vehicle is 60 km/h and the centrifugal ratio for the road is $1/4$, the minimum radius of the curve in m is

- (A) 113.26 (B) 98.18 (C) 25.46 (D) 15.50

Answer (A)

Solution .

We have,

$$\text{Centrifugal ratio} = \frac{P}{W} = \frac{v^2}{gR}$$

Where, P – centrifugal force

W – weight of vehicle

$$\therefore \frac{1}{4} = \frac{(16.67)^2}{9.81 \times R}$$

$$\therefore R = 113.26 \text{ m}$$

Q.12 A centrifugal fan rotating at 500 rpm delivers 70 m³/s of air. If the speed is reduced to 200 rpm,

The quantity of air delivered in m³/s will be

- (A) 175 (B) 55 (C) 28 (D) 11

Answer (C)

Solution .

We know, Quantity of air (Q) delivered varies directly as fan-speed (N).

$$Q \propto N$$

$$\text{Or } \frac{Q_1}{Q_2} = \frac{N_1}{N_2}$$

$$\therefore \frac{70}{Q_2} = \frac{500}{200}$$

$$\therefore Q_2 = 28 \text{ m}^3/\text{sec}$$

Q 13. According to mine regulations, the value of the fleet angle α , in degree of a drum winder installation lies in the range of

- (A) $1.5 < \alpha \leq 2.0$ (B) $0 < \alpha \leq 1.5$ (C) $2.0 < \alpha \leq 2.5$ (D) $2.5 < \alpha \leq 3.0$

Answer (B)

Q.14 Water will not be delivered by a centrifugal pump due to

- (A) lack of priming (B) too low discharge head
(C) wrong direction of rotation (D) partial obstruction at discharge outlet

Answer (A)

Q.15 Match the following

Mine car type	Mode of unloading
P. Granby	1. Bottom opening
Q. Gable bottom	2. Both side tilting
R. Drop bottom	3. Single side opening
S. Rocker dump	4. Both side opening
(A) P-2, Q-4, R-3, S-1	(B) P-4, Q-1, R-3, S-2
(C) P-3, Q-1, R-4, S-2	(D) P-3, Q-4, R-1, S-2

Answer (D)

Q.16 Mean air temperature of a 450 m deep downcast shaft is 29°C and that of the upcast shaft is 37°C. The height of the motive column in m is

- (A) 8.2 (B) 9.5 (C) 11.6 (D) 12.8

Answer (C)

Solution .

We have,

$$\text{Motive column} = D \times \frac{T_u - T_d}{273 + T_u}, \text{ m}$$

$$\therefore \text{Motive column} = 450 \times \frac{37-29}{273+37}, \text{ m}$$

$$\therefore \text{Motive column} = 11.6 \text{ m}$$

Q.17 The total pressure and the static pressure measured at a point in a ventilation duct are 20 mm and 10 mm of water gauge respectively. If density of air is 1.2 kg/m³, the velocity of the air in m/s is

(A) 14.08

(B) 12.78

(C) 8.53

(D) 6.24

Answer (B)

Solution :

We know,

Total pressure = velocity pressure + static pressure

$$20 = 10 + \text{velocity pressure}$$

Velocity pressure = 10 mm of w.g

Or velocity pressure = 100 Pa

$$\text{Or velocity pressure} = \frac{1}{2} \times \rho \times v^2$$

Where, ρ - density of air in $\text{kg/m}^3 = 1.2 \text{ kg/m}^3$

v- velocity of air in m/sec

$$100 = \frac{1}{2} \times \rho \times v^2$$

$$\text{Or } v^2 = \frac{2 \times 100}{1.2}$$

$$\therefore v = 12.93 \text{ m/sec}$$

Q.18 The type of fire extinguisher that must NOT be used in case of fire in an electric substation located in an underground metal mine is

(A) multi-purpose dry chemical extinguisher

(B) CO_2 snow extinguisher

(C) dry chemical powder extinguisher

(D) foam extinguisher

Answer (D)

Q.19 ISO 9000 Quality Systems DO NOT contain

(A) legal provisions

(B) measurement

(C) document control

(D) standardization

Answer (A)

- Q.20** Air samples collected from the intake and the return gates of a retreating longwall face show methane concentration values of 0.1 % and 0.8 % respectively. The production from the longwall face is 2000 tonne/day and the air quantity circulating the face is 15 m³/s. The rate of methane emission in m³ per tonne of coal produced is
- (A) 11.0 (B) 9.5 (C) 5.5 (D) 4.5

Answer (D)

Solution .

$$\text{Quantity of methane picked up by air} = \frac{15 \times (0.8 - 0.1)}{100}$$

$$= 0.105 \text{ m}^3/\text{sec}$$

$$= 9072 \text{ m}^3/\text{day}$$

$$\therefore \text{Emission of methane per tonne of coal} = \frac{9072}{2000} = 4.5 \text{ m}^3/\text{tone}$$

- Q.21** The time study data of an equipment deployed in a mine during a calendar month is given below.

Total working hours = 400

Total maintenance hours = 100

Total hours of actual work = 240

The percentage of utilization of the equipment is

- (A) 85 (B) 80 (C) 65 (D) 60

Answer (B)

Solution .

$$\text{Percentage of utilization of equipment} = \frac{240}{400 - 100} \times 100$$

$$= 80 \%$$

- Q.22 100 ml of waste water is allowed to evaporate in a dish weighing 48.6232 g. The weight of the dish with dry solids is 48.6432 g. The concentration of dry solids in waste water in mg/l is
- (A) 200 (B) 220 (C) 260 (D) 320

Answer (A)

Solution

$$\begin{aligned}\text{Weight of dry solids in 100 ml waste water} &= 48.6432 - 48.6232 \\ &= 0.02 \text{ g}\end{aligned}$$

So, 100 ml contains 0.02 g dry solids

$$1 \text{ ml contains} = \frac{0.02}{100} \text{ g dry solids}$$

$$\text{And 1 l waste water contains} = \frac{0.02 \times 1000}{100} \text{ g dry solids}$$

$$\text{Or 1 l waste water contains} = \frac{0.02 \times 1000 \times 1000}{100} \text{ mg dry solids}$$

$$\text{Or 1 l waste water contains} = 200 \text{ mg dry solids}$$

Hence, concentration of dry solids in waste water is 200 mg/L.

- Q.23 A longwall face cut by double back shuffle method can be only worked with
- (A) fixed drum shearer (B) single ended ranging drum shearer
- (C) double ended ranging drum shearer (D) plough

Answer (C)

- Q.24 For an oil exploration drilling, chance of striking an oil reservoir is 1 out of 15. If an oil exploration company decides to explore 5 sites, the probability of striking at least one successful oil reservoir is

(A) 0.292

(B) 0.250

(C) 0.034

(D) 0.0024

Answer (A)

Solution .

Let $P(A) = 1/15$; $P(B) = 1/15$; $P(C) = 1/15$; $P(D) = 1/15$; $P(E) = 1/15$;

The probability of striking at least one successful oil reservoir is-

$$P = 1 - P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) \cdot P(\bar{D}) \cdot P(\bar{E})$$

$$P = 1 - \{(1 - P(A))\}\{(1 - P(B))\}\{(1 - P(C))\}\{(1 - P(D))\}\{(1 - P(E))\}$$

$$P = 1 - (1 - 1/15) \cdot (1 - 1/15) \cdot (1 - 1/15) \cdot (1 - 1/15) \cdot (1 - 1/15)$$

$$\therefore P = 0.292$$

Q.25 Product of the eigen values of the matrix $A = \begin{bmatrix} 3 & 2 & 5 \\ 2 & 2 & 1 \\ 1 & 5 & 4 \end{bmatrix}$ is

(A) 6

(B) 8

(C) 10

(D) 35

Answer (D)

Solution .

We know,

The product of eigen value of a given matrix is equal to the determinant of given matrix.

$$A = \begin{bmatrix} 3 & 2 & 5 \\ 2 & 2 & 1 \\ 1 & 5 & 4 \end{bmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 3 & 2 & 5 \\ 2 & 2 & 1 \\ 1 & 5 & 4 \end{vmatrix}$$

$$|A| = 3 \begin{vmatrix} 2 & 1 \\ 5 & 4 \end{vmatrix} - 2 \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} + 5 \begin{vmatrix} 2 & 2 \\ 1 & 5 \end{vmatrix}$$

$$|A| = 3(8 - 5) - 2(8 - 1) + 5(10 - 2)$$

$$|A| = 35$$

Q.26 For the equation $\frac{dy}{dx} = 2x + 3y$, the value of y at $x = 0.1$ in one step using Runge-Kutta

Fourth order method for the condition $y = 1$ when $x = 0$, is

- (A) 0.3608 (B) 1.2508 (C) 1.3608 (D) 1.4625

Answer (C)

Solution .

$$f(x, y) = 2x + 3y, \quad x_0 = 0, y_0 = 1, h = 0.1 - 0 = 0.1$$

$$\text{Now, } K_1 = h \cdot f(x_0, y_0) = 0.1 \times (2x_0 + 3y_0)$$

$$= 0.3$$

$$K_2 = h \cdot f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1 f(0.05, 1.15)$$

$$= 0.355$$

$$K_3 = h \cdot f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1 f(0.05, 1.1775)$$

$$= 0.36325$$

$$K_4 = h \cdot f(x_0 + h, y_0 + k_3) = 0.1 f(0.1, 1.36325)$$

$$= 0.428975$$

$$\text{Now, } k = \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}$$

$$\therefore k = 0.3609125$$

$$\text{Hence, Value of } y = 1 + 0.3609125 = 1.3609125$$

Q.27 The Value of the integral $\int_0^1 \sqrt{\frac{1+x}{1-x}} dx$ is

- (A) $\frac{\pi}{2} - 1$ (B) $\frac{\pi}{2} + 1$ (C) $\pi - 1$ (D) $\pi + 1$

Answer (B)

Solution .

$$I = \int_0^1 \sqrt{\frac{1+x}{1-x}} dx$$

$$I = \int_0^1 \sqrt{\frac{1+x}{1-x}} \times \frac{1+x}{1+x} dx$$

$$I = \int_0^1 \frac{1+x}{\sqrt{1-x^2}} dx$$

$$I = \int_0^1 \frac{1}{\sqrt{1-x^2}} dx + \int_0^1 \frac{x}{\sqrt{1-x^2}} dx$$

$$I = [\sin^{-1} x]_0^1 + \int_0^1 \frac{x}{\sqrt{1-x^2}} dx$$

$$\text{Let, } 1 - x^2 = t$$

$$\therefore dx = \frac{-dt}{2x}$$

When $x = 1$ then $t = 0$ & When $x = 0$ then $t = 1$,

$$I = [\sin^{-1} 1 - \sin^{-1} 0] - \int_1^0 \frac{x}{\sqrt{t}} \cdot \frac{dt}{2x}$$

$$\therefore I = \frac{\pi}{2} - \frac{1}{2} \left[\frac{t^{1/2}}{1/2} \right]_1^0$$

$$\therefore I = \frac{\pi}{2} + 1$$

Q.28 A 1 tonne mine car traveling at a constant speed of 10 km/h collides with a stationary buffer and comes to rest. If the buffer spring stiffness is 200 kN/m, the maximum compression in the spring in mm is

(A) 49

(B) 98

(C) 196

(D) 247

Answer (C)

Solution .

$$\text{Kinetic energy } (E_K) \text{ of mine car} = \frac{1}{2} mv^2$$

$$\text{Where, } m = \text{mass of mine car} = 1 \text{ tonne} = 10^3 \text{ kg}$$

$$v = \text{speed of mine car} = 10 \text{ km/h} = 2.78 \text{ m/sec}$$

$$\text{Potential energy } (E_P) \text{ of spring} = \frac{1}{2} kx^2$$

$$\text{where, } k = \text{spring stiffness} = 200 \text{ kN/m} = 200 \times 10^3 \text{ N/m}$$

x = spring displacement or compression

$$\therefore E_K = E_P$$

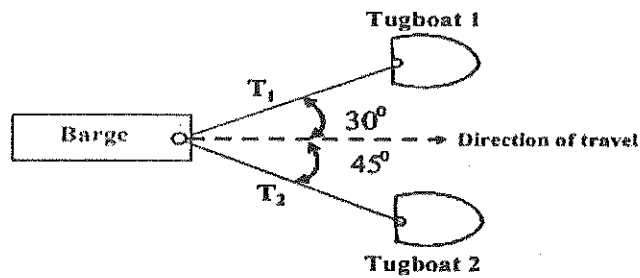
$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$

$$x = \sqrt{\frac{mv^2}{k}}$$

$$\therefore x = \sqrt{\frac{10^3 \times 2.78^2}{200 \times 10^3}}$$

$$\therefore x = 0.196 \text{ m} = 196 \text{ mm}$$

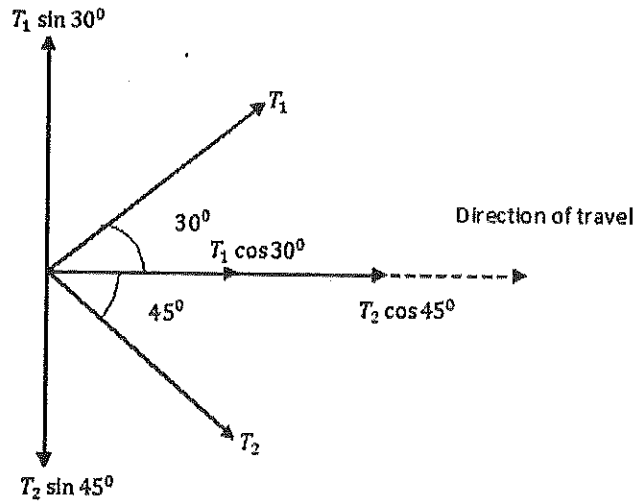
- Q.29** In an iron ore handling port, a barge is pulled by ropes using two tugboats as shown in the figure. In equilibrium, the resultant of the forces T_1 and T_2 along the axis of the barge in the direction of its travel is 5000 N. The tensions T_1 and T_2 in N respectively are



- (A) 9700 and 6831 (B) 6831 and 9700
(C) 3660 and 2588 (D) 2588 and 3660

Answer (C)

solution .



$$\therefore \sum V = 0$$

$$\therefore T_1 \sin 30^\circ - T_2 \sin 45^\circ = 0 \dots\dots\dots(1)$$

$$\therefore \sum H = 0$$

$$\therefore T_1 \cos 30^\circ + T_2 \cos 45^\circ = 5000 \dots\dots\dots(2)$$

From equation (1) & (2), it is found that

$$\therefore T_1 = 3660 \text{ N} \text{ \& } T_2 = 2588 \text{ N}$$

Q.30 A flat belt conveyor is carrying coal of bulk density 1 tonne/m^3 at a rate of 400 tonne/h . The belt speed is 3 m/s . Coal is spread over the belt covering 80% of the belt width in a shape of a triangle. If the pile height is $1/4$ of the belt width, the width of the belt in mm is

- (A) 1109 (B) 909 (C) 709 (D) 609

Answer (D)

Solution .

Given that , pile height = $\frac{\text{belt width}}{4}$

Productivity of belt conveyor (T) = 400 tonne/hour

$$T = \frac{400}{60 \times 60} \text{ tonne/sec}$$

And we have,

$$T = a . b . v \text{ tonne/sec}$$

$$\text{Or } T = \left(\frac{1}{2} \times \text{width of belt} \times \text{pile height} \right) \times b \times v$$

$$\text{Or } T = \left(\frac{1}{2} \times \text{width of belt} \times \frac{\text{belt width}}{4} \right) \times b \times v$$

$$\text{Or } T = \left(\frac{1}{2} \times \frac{(\text{belt width})^2}{4} \right) \times b \times v$$

Now,

$$\frac{400}{60 \times 60} = \left(\frac{1}{2} \times \frac{(\text{belt width})^2}{4} \right) \times 1 \times 3$$

$$\therefore \text{belt width} = 609 \text{ mm}$$

Q.31 Match the following.

Hydraulic system components

Symbols

P. Fixed displacement unidirectional flow pump

1.



Q. Fixed displacement unidirectional flow motor

2.



R. Accumulator

3.



S. Filter

4.



(A) P-4, Q-2, R-3, S-1

(B) P-2, Q-4, R-3, S-1

(C) P-3, Q-2, R-1, S-4

(D) P-2, Q-3, R-1, S-4

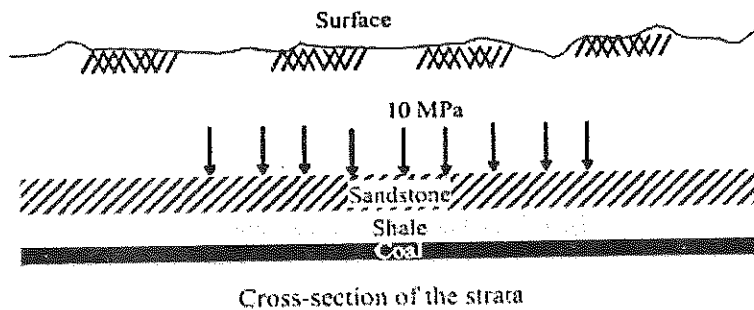
Answer (B)

Q.32 Match the following

Method of mining	Stope support	Ore loading
P. Shrinkage stoping	1. Insitu pillar	a. Overhead mucker
Q. Blasthole stoping	2. Broken ore	b. Pneumatic autoloader
R. Top slicing	3. Timber mat	c. Load haul dumper
(A) P-2-a, Q-1-c, R-3-b		(B) P-2-a, Q-3-c, R-1-b
(C) P-2-b, Q-3-c, R-1-a		(D) P-3-c, Q-2-a, R-1-b

Answer (A)

Q.33 A typical case of gravity loading under complete lateral restraint in flat strata is shown in the figure below. The physico-mechanical parameters of the strata are given in the table. The in situ stresses (σ_z , σ_H) on the top of the coal seam in MPa are



Strata	Thickness (m)	Specific Gravity	Young's Modulus (GPa)	Shear Modulus (GPa)
Sandstone	50	2.35	26.40	12.5
Shale	25	2.15	20.50	8.25
Coal	20	1.52	2.41	0.95

- (A) (10.17, 2.54) (B) (10.17, 3.69) (C) (11.68, 3.69) (D) (11.68, 2.54)

Answer (C)

Solution .

Vertical stress:-

$$\sigma_v = \gamma \cdot z$$

$$\text{Or } \sigma_v = \rho \cdot g \cdot z$$

Where,

ρ - density of rock

g - acceleration due to gravity

z - depth below surface

γ - unit weight of rock

Now, vertical stress due to sandstone on the top of the shale:-

$$\sigma_v' = 2.35 \times 1000 \times 9.81 \times 50 \text{ N/m}^2$$

$$\sigma_v' = 1.15 \text{ MPa}$$

Vertical stress due to shale on the top of coal:-

$$\sigma_v'' = 2.15 \times 1000 \times 9.81 \times 25 \text{ N/m}^2$$

$$\sigma_v'' = 0.527 \text{ MPa}$$

$\therefore$ vertical stress(σ_z) on the top of the coal seam due to overlying strata :-

$$\sigma_z = 10 + 1.15 + 0.527 = 11.68 \text{ MPa}$$

Given for shale :-

$$G = \text{shear modulus} = 8.25 \text{ GPa}$$

$$E = \text{young modulus} = 20.50 \text{ GPa}$$

And we know,

$$G = \frac{mE}{(m+1)}$$

Where, m - Poisson number ;

$$\therefore 8.25 = \frac{m \times 20.50}{2(m+1)}$$

$$\therefore m = 4.125$$

$$\therefore \text{poisson ratio} = \nu = \frac{1}{m} = 0.24$$

$$\text{And } k = \frac{\nu}{1-\nu} = \frac{0.24}{1-0.24} = 0.315$$

$$\therefore \text{Horizontal stress } (\sigma_H) = k \cdot \sigma_z$$

$$\sigma_H = 0.315 \times 11.68$$

$$\sigma_H = 3.69 \text{ MPa}$$

The sale value of chromite ore from an open pit mine is Rs. 6500 per tonne. Cost of mining, excluding stripping cost, is Rs. 2450 per tonne. If the cost of stripping is Rs. 1150 per m^3 , the breakeven stripping ratio in m^3/tonne is

- (A) 2.18 (B) 3.52 (C) 3.65 (D) 4.25

Answer (B)

tion.

break even stripping ration

$$\begin{aligned} &= \frac{\text{Recoverable price per tonne of ore} - \text{O.C. production cost per tonne of ore}}{\text{cost of extraction per cubic meter of overburden material}} \\ &= \frac{(6500) - (2450)}{1150} \text{ m}^3/\text{tonne} \\ &= 3.52 \text{ m}^3/\text{tonne} \end{aligned}$$

An investment at 10% yearly interest rate, compounded quarterly, accumulates to a sum of Rs. 120,000 in 5 years. The present value of the sum in rupees is

- (A) 73,233 (B) 74,511 (C) 88,232 (D) 106,063

Answer (A)

tion.

Relation between Future Value (FV) and Present Value (PV) is given by-

$$FV = PV \left(1 + \frac{r}{n}\right)^{nt}$$

Where,

r- rate of interest

t- times in years

n - number of times per year, interest is compounded

Therefore,

$$120000 = PV \left(1 + \frac{0.1}{4}\right)^{4 \times 5}$$

$$\therefore PV = 73232 \text{ rupees}$$

Q.36 A toxic gas flows into a mine working place at the rate of $2.52 \text{ m}^3/\text{min}$. The concentration of the gas in the intake air is 0.25%. The minimum quantity of intake air in m^3/min required to dilute the gas to its threshold limit value of 1.0 % is

(A) 123

(B) 252

(C) 295

(D) 333

Answer (D)

Solution .

$$Q = \frac{100 q}{C_p - C_a} - q$$

$$Q = \frac{100 \times 2.52}{1 - 0.25} - 2.52$$

$$Q = 333 \text{ m}^3/\text{min}.$$

Q.37 An exhaust fan attached to an evasee of 18 m^2 cross-sectional area at the outlet circulates $150 \text{ m}^3/\text{s}$ of air at the pressure of 1000 Pa in a mine ventilation system. The ratio of the inlet to outlet area of the evasee is 1:4 and the density of air is 1.2 kg/m^3 . The quantity of air circulated in the mine in absence of evasee is $120 \text{ m}^3/\text{s}$. The evasee efficiency in % is

(A) 57.6

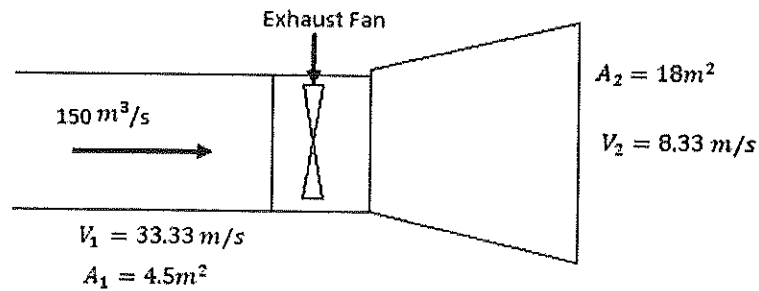
(B) 43.2

(C) 39.06

(D) 37.7

Answer (A)

Solution .



$$\therefore \frac{A_1}{A_2} = \frac{1}{4}$$

$$\therefore A_1 = \frac{18}{4} = 4.5 \text{ m}^2$$

And $V_1 = \frac{Q}{A_1} = \frac{150}{4.5} = 33.33 \text{ m/sec}$

$$V_2 = \frac{Q}{A_2} = \frac{150}{18} = 8.33 \text{ m/sec}$$

$$\begin{aligned} \text{Pressure recovery by evasee} &= \frac{1}{2} \times \rho \times (V_1^2 - V_2^2) \times \text{recovery efficiency}(\eta), \text{ Pa} \\ &= \frac{1}{2} \times 1.2 \times (33.33^2 - 8.33^2) \times \eta \\ &= 624.9 \times \eta \text{ Pa} \end{aligned}$$

Now, In absence of evasee-

$$(1000 - 624.9 \times \eta) = R \times 120^2 \dots\dots\dots(1)$$

In presence of evasee-

$$1000 = R \times 150^2 \dots\dots\dots(2)$$

From equation (1) & (2),

$$\frac{(1000 - 624.9 \times \eta)}{120^2} = \frac{1000}{150^2}$$

$$\therefore \eta = \frac{1000 - 640}{624.9}$$

$$\therefore \eta = 57.6 \%$$

- Q.38 A fan circulates $24 \text{ m}^3/\text{s}$ of air at a pressure of 1200 Pa in a ventilation district. It is intended to reduce the air quantity to $16 \text{ m}^3/\text{s}$ by placing a regulator. Assuming the pressure remains unchanged, the size of the regulator in m^2 is
- (A) 1.48 (B) 0.74 (C) 0.37 (D) 0.18

Answer (B)

Solution .

We know,

$$P = RQ^2$$

Before installation of regulator-

$$1200 = R \times 24^2$$

$$\therefore R = 2.08 \text{ N s}^2 \text{ m}^{-8}$$

After installation of regulator-

$$P = (R + R_R) Q^2$$

Where , R_R – resistance of regulator

$$1200 = (2.08 + R_R) \times 16^2$$

$$R_R = \frac{1200}{256} - 2.08$$

$$R_R = 2.608 \text{ N s}^2 \text{ m}^{-8}$$

$$\begin{aligned} \text{Now, size of regulator} &= \frac{1.2}{\sqrt{R_R}} \\ &= \frac{1.2}{\sqrt{2.608}} \\ &= 0.74 \text{ m}^2 \end{aligned}$$

- Q.39 An air sample taken from the return airway of a district contains the following gases. The Graham's ratio for the district is

Gas	Concentration (%)
CO_2	0.40
H_4	1.17
O_2	19.92
N_2	78.49
CO	0.02

(A) 5.6

(B) 4.8

(C) 3.0

(D) 2.3

Answer (D)

Solution .

Atmosphere air constituents following which goes down the mine (by volume %) :-

Oxygen - 20.93 %

Nitrogen (including Argon and other rare inert gas) - 79.04 %

Carbon dioxide – 0.03 %

Now, percentage of oxygen, when percentage of nitrogen is 78.49 % = $\frac{20.93}{79.04} \times 78.49$

$$= 20.79 \%$$

$$\therefore \text{oxygen absorbed} = 20.79 - 19.92$$

$$= 0.87 \%$$

$$\text{Now, Graham's ratio} = \frac{CO \text{ produced}}{O_2 \text{ consumed}} \times 100$$

$$= \frac{0.02}{0.87} \times 100$$

$$= 2.3 \%$$

Q.40 An incandescent headlight of a mining vehicle is of spot beam type with a beam angle of 30° . The spherical surface in m^2 subtended by the lighted beam at a distance of 5 m from the headlight is

(A) 7.5

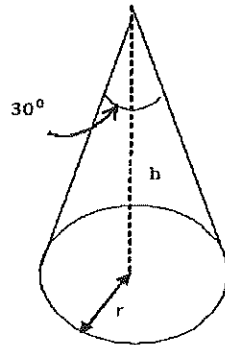
(B) 15

(C) 21

(D) 25

Answer (C)

Solution .



Spot beam angle = 30^0

Spot area = πr^2

But in case, of sphere-

Spherical surface area = solid angle $\times r^2$

Solid angle = $2\pi(1 - \cos 30^0)$

$\therefore$ solid angle = 0.841

Therefore, spherical surface area = $0.841 \times 5^2 = 21 \text{ m}^2$

Q.41 The total cost C (lakh rupees) of a longwall face of length L in m is given by the equation

$C = 0.1L + \frac{1562.5}{L} + 300$. Length of the face in m for the minimum total cost is

(A) 40

(B) 125

(C) 156

(D) 300

Answer (B)

Solution .

$$c = 0.1 L + \frac{1562.5}{L} + 300 \dots\dots\dots(1)$$

Differentiating equation (1), w.r.t ,L

$$\frac{dc}{dL} = 0.1 - \frac{1562.5}{L^2}$$

But for minimum or maximum

$$\frac{dc}{dL} = 0$$

$$0.1 - \frac{1562.5}{L^2} = 0$$

$$\therefore L = \sqrt{\frac{1562.5}{0.1}}$$

$$\therefore L = 125 \text{ m}$$

Q.42 20 plain detonators in series, each of 2Ω resistance, are fired by a DC exploder supplying a current of 1.25 A. If 250 mJ energy is spent to fire the detonators, the time required in millisecond after detonator initiation is

(A) 4

(B) 8

(C) 12

(D) 16

Answer (A)

Solution .

$$\text{Electric power (P)} = i^2 \times R$$

$$P = 1.25^2 \times (2 \times 20)$$

$$P = 62.5 \text{ W}$$

$$\text{And Power(P)} = \frac{\text{work(w)}}{\text{time(t)}}$$

$$\text{Or } w = P.t$$

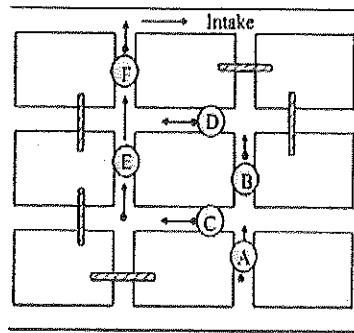
$$250 = 62.5 \times t$$

$$\therefore t = \frac{250}{62.5} = 4 \text{ ms}$$

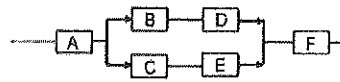
Q.43 A sudden increase of CO incidence has occurred in an underground mine section. A man at point A starts to run out to the main intake of the mine where he will be safe. Refer figure

below for the mine section and the logic diagram. The probabilities that he will successfully cross the gallery sections A, B, C, D, E, and F are 0.9, 0.8, 0.7, 0.8, 0.7 and 0.9 respectively.

The probability that he will successfully reach the main intake is



Section of the mine



Series-parallel logic diagram

(A) 0.40

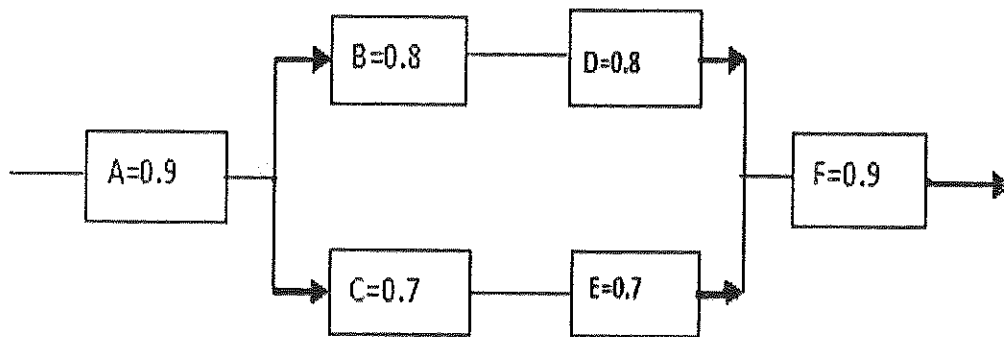
(B) 0.51

(C) 0.66

(D) 0.77

Answer (C)

Solution .



For sections B&D -

Reliability will be $= 0.8 \times 0.8 = 0.64$

For sections C&E-

Reliability will be $= 0.7 \times 0.7 = 0.49$

Now, Reliability for sections BD & CE will be-

$$= 1 - (1 - 0.64) \times (1 - 0.49)$$

$$= 0.8164$$

Hence,

$$\text{Required probability} = 0.9 \times 0.9 \times 0.8164$$

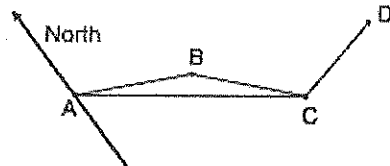
$$= 0.66$$

Q.44 In an underground correlation survey by the Weisbach triangle (figure below) the following data are obtained.

AB = 3.50 m, BC = 2.75 m, CA = 6.20 m,

Angle $\widehat{ACD} = 179^\circ 14'33''$, Angle $\widehat{BCD} = 179^\circ 10'17''$ and

bearing of AB = $115^\circ 23'49''$. The bearing of traverse CD is



(A) $102^\circ 27'16''$

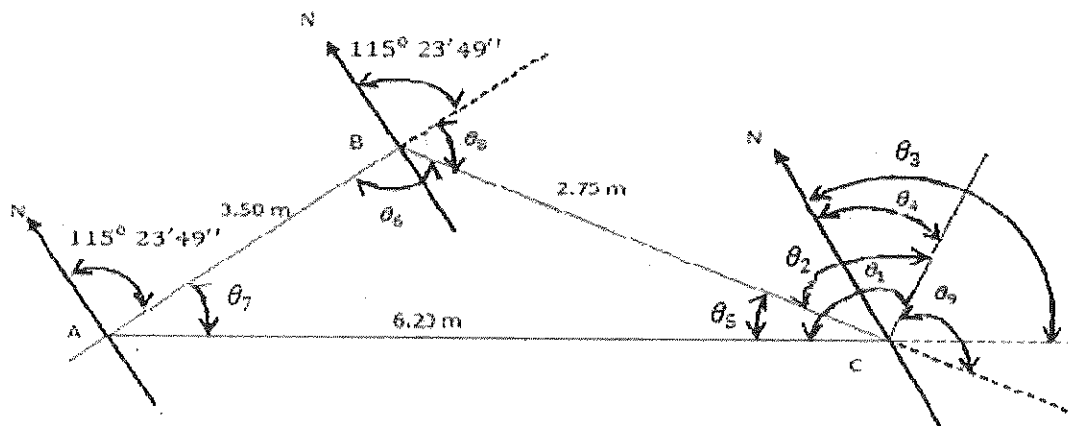
(B) $114^\circ 41'49''$

(C) $115^\circ 27'16''$

(D) $179^\circ 14'16''$

Answer (B)

Solution .



$$\therefore \theta_1 = 179^0 14' 33''$$

$$\theta_2 = 179^0 10' 17''$$

$$\therefore \theta_5 = \theta_1 - \theta_2 = 0^0 4' 16''$$

Applying sine formula in ΔABC

$$\frac{3.50}{\sin \theta_5} = \frac{2.75}{\sin \theta_7}$$

$$\therefore \theta_7 = 0^0 3' 21.14''$$

So,

$$\theta_6 = 180^0 - \theta_5 - \theta_7$$

$$\therefore \theta_6 = 179^0 52' 86''$$

$$\theta_8 = 180^0 - \theta_6$$

$$\therefore \theta_8 = 0^0 7' 37.14''$$

$$\theta_3 = 115^0 23' 49'' + 0^0 7' 37.14'' = 115^0 31' 26.14''$$

$$\theta_9 = 180^0 - \theta_2 = 0^0 49' 43''$$

Therefore, bearing of CD-

$$\theta_4 = \theta_3 - \theta_9$$

$$\theta_4 = 114^0 41' 49''$$

Common Data for Questions 45 and 46:

A concentrator pilot plant is fed with 1 tonne of copper ore at ROM grade of 1.5 % Cu. Metal recovery in the concentrator pilot plant is 90% and the grade of copper in concentrate is 20%.

Q 45. The amount of copper in concentrate in kg is

(A) 13.5

(B) 14.0

(C) 14.5

(D) 15.0

Answer (A)

Solution 48.

Grade of copper = 1.5 % (ROM)

Copper content per tonne of ore = $\frac{1.5}{100} \times 1000 = 1.5 \text{ kg}$

Copper recovered by mill = $15 \times \frac{90}{100} = 13.5 \text{ kg}$

Hence, amount of copper in concentrate = 13.5 kg of copper

Q.46 Amount of concentrate produced from 1 tonne of ore in kg is

(A) 75.0

(B) 72.0

(C) 70.0

(D) 67.5

Answer (D)

Solution .

Grade of copper in concentrate = 20 %

It means that,

100 kg of concentrate has 20 kg of copper

Or 1 kg of copper produced from $\frac{100}{20}$ kg of concentrate.

Hence,

Amount of concentrate produced from 1 tonne of ore:-

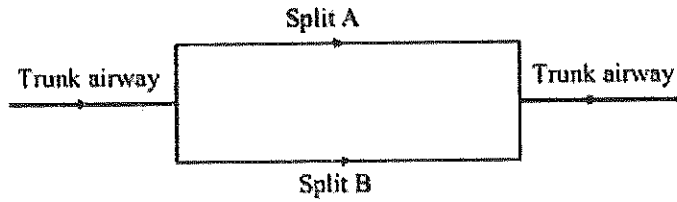
$$= 13.5 \times \frac{100}{20} \text{ kg of concentrate}$$

$$= 67.5 \text{ kg of concentrate}$$

Common Data for Questions 47 and 48:

A mine ventilation system consists of two splits A and B with resistances of $0.8 \text{ N s}^2 \text{ m}^{-8}$ and $3.2 \text{ N s}^2 \text{ m}^{-8}$, respectively as shown in figure. Trunk airways have resistance of $0.2 \text{ N s}^2 \text{ m}^{-8}$.

The main mine fan is generating pressure of 500 Pa.



Q.47. The air quantities in m^3/s circulated in the splits A and B respectively are

(A) 20 and 30

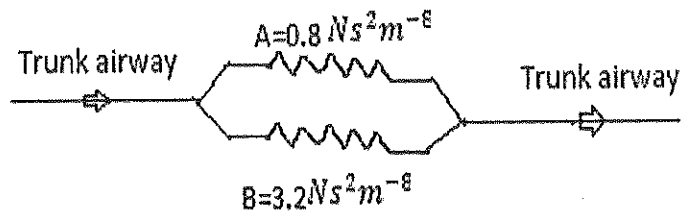
(B) 30 and 20

(C) 20 and 10

(D) 40 and 10

Answer (C)

Solution .



$$\frac{1}{\sqrt{R_{AB}}} = \frac{1}{\sqrt{R_A}} + \frac{1}{\sqrt{R_B}}$$

$$\frac{1}{\sqrt{R_{AB}}} = \frac{1}{\sqrt{0.8}} + \frac{1}{\sqrt{3.2}}$$

$$\therefore R_{AB} = 0.36 \text{ N s}^2 \text{ m}^{-8}$$

$$\text{Total resistance} = 0.20 + 0.36 = 0.56 \text{ N s}^2 \text{ m}^{-8}$$

$$\therefore P = R_T Q_T^2$$

$$500 = 0.56 \times Q_T^2$$

$$\therefore Q_T = 30 \text{ m}^3/\text{sec}$$

fan Pressure = pressure lost in trunk roadway + pressure lost in split A

$$500 = R_{T_r} Q_T^2 + R_A Q_A^2$$

$$500 = 0.2 \times 30^2 + 0.8 \times Q_A^2$$

$$\therefore Q_A = 20 \text{ m}^3/\text{sec}$$

$$\text{And, } Q_B = Q_T - Q_A = 30 - 20 = 10 \text{ m}^3/\text{sec}$$

Q.48 The flows in the two splits are equalized by placing a booster fan in split B. Assume that the fan pressure does not change after installation of the booster fan. The size of the booster fan in Pa is

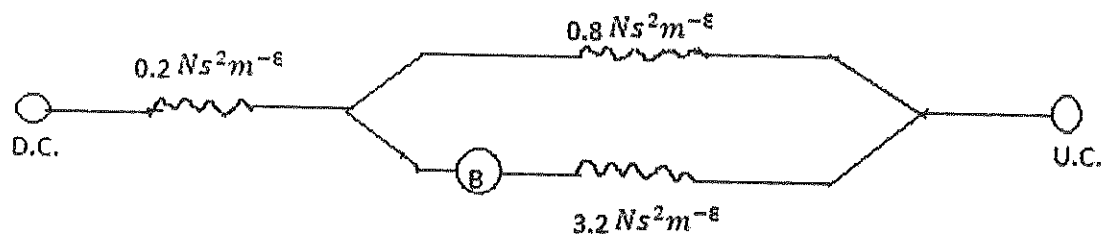
(A) 749.05

(B) 850.08

(C) 950.02

(D) 1000.50

Solution :



After installation of booster fan:-

Fan Pressure = pressure lost in trunk airways + pressure lost in split A

$$500 = R_{T_r} Q_T^2 + R_A Q_A^2$$

$$500 = 0.2 \times Q_T^2 + 0.8 \times (Q_T - Q_B)^2$$

$$500 = 0.2 \times Q_T^2 + 0.8 \times (Q_T - 20)^2 \quad [\because Q_B = 20 \text{ m}^3/\text{sec}]$$

$$\therefore Q_T = 36.88 \text{ m}^3/\text{sec}$$

Now, flow in split B

$$500 + P_B = R_{T_r} Q_T^2 + R_B Q_B^2$$

Where, P_B = booster fan pressure

$$500 + P_B = 0.2 \times 36.88^2 + 3.2 \times 20^2$$

$$\therefore P_B = 1052 \text{ Pa}$$

Statement for Linked Answer Questions 49 and 50:

A 400 V, 3 phase, star connected induction motor takes a line current of 10 A with 0.86 p.f. lagging. Total stator losses are 5 % of the input. Rotor copper losses are 4 % of the input to the rotor, and mechanical losses are 3 % of the input to the rotor.

- Q.49 The input power to the rotor in Watts is
- (A) 5958 (B) 5788 (C) 5660 (D) 5532

Answer (C)

Solution .

We have ,

$$\text{Input power (P)} = \sqrt{3} \cdot V \cdot I \cdot \cos \phi$$

$$P = \sqrt{3} \times 400 \times 10 \times 0.86$$

$$P = 5958.25 \text{ W}$$

Given that , total stator losses = 5 % of input power

Hence, input power to the rotor is:-

$$P = 5958.25 - 5958.25 \times \frac{5}{100}$$

$$P = 5660.34 \text{ W}$$

- Q.50 The shaft output power in Watts is
- (A) 5562 (B) 5490 (C) 5434 (D) 5264

Answer (D)

Solution .

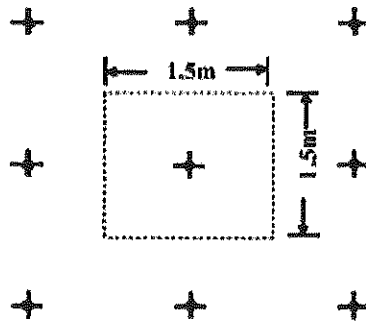
Given that total copper losses and mechanical losses are 4% and 3% of input power respectively.

$$\text{So shaft power output} = 5660.34 - 5660.34 \times \frac{4}{100} - 5660.34 \times \frac{3}{100}$$

$$= 5264.118 \text{ watt}$$

Statement for Linked Answer Questions 51 and 52:

The bolts are spaced at 1.5 m centre-to-centre in a square pattern as shown in the figure below. The tensile stress in 22 mm diameter bolt rod is 193.35 MPa. The unit weight of the roof layer is 25 kN/m³.



Plan view of rock bolting pattern

Q.51 The axial load in the bolt rod in kN is

- (A) 294.0 (B) 173.5 (C) 147.0 (D) 73.5

Answer (D)

Solution .

$$\text{stress} = \frac{\text{load}}{\text{area}}$$

$$\text{Or tensile stress} = \frac{\text{axial load}}{\text{area}}$$

$$\text{Or axial load} = \text{area} \times \text{tensile stress}$$

$$\begin{aligned} \text{axial load} &= 3.14 \times \frac{(22 \times 10^{-3})^2}{4} \times 193.35 \times 10^6 \\ &= 73461 \text{ N} \\ &= 73.461 \text{ kN} \end{aligned}$$

Q.52 At equilibrium, the thickness of the roof layer supported by the bolt in m is

- (A) 1.31 (B) 2.4 (C) 2.62 (D) 3.08

Answer (A)

Solution .

We have,

$$\text{Unit weight of layer} = \frac{\text{weight of layer}}{\text{volume of layer}}$$

$$25 = \frac{73.5}{1.5 \times 1.5 \times \text{thickness of roof layer}}$$

$$\therefore \text{thickness of roof layer} = \frac{73.5}{1.5 \times 1.5 \times 25}$$

$$\therefore \text{thickness of roof layer} = 1.31 \text{ m}$$

Q.53 Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Deference

(A) aversion (B) resignation (C) suspicion (D) contempt

Answer (D)

Q.54 Choose the most appropriate word(s) from the options given below to complete the following sentence.

We lost confidence in him because he never _____ the grandiose promises he had made.

(A) delivered (B) delivered on (C) forgot (D) reneged on

Answer (B)

Q.55 Choose the word or phrase that best completes the sentence below.

_____ in the frozen wastes of Arctic takes special equipment.

- (A) To survive (B) Surviving (C) Survival (D) That survival

Answer (A)

- Q.56** In how many ways 3 scholarships can be awarded to 4 applicants, when each applicant can receive any number of scholarships?

- (A) 4 (B) 12 (C) 64 (D) 81

Answer (C)

Solution .

Number of ways in which 3 scholarship can be awarded to 4 applicant, when each receive any number of scholarship = $4^3 = 64$

- Q.57** Choose the most appropriate word from the options given below to complete the following sentence.

The _____ of evidence was on the side of the plaintiff since all but one witness testified that his story was correct.

- (A) paucity
(B) propensity
(C) preponderance
(D) accuracy

Answer (C)

- Q.58** If $(2y+1)/(y+2) < 1$, then which of the following alternatives gives the CORRECT range of y?

- (A) $-2 < y < 2$ (B) $-2 < y < 1$ (C) $-3 < y < 1$ (D) $-4 < y < 1$

Answer (B)

Solution .

$$\frac{2y+1}{y+2} < 1 \dots\dots\dots(1)$$

Put , $y = -1$, to satisfy equation (1)

$$\frac{-2+1}{-1+2} < 1$$

$$-1 < 1$$

Now,

$$\text{Case1} \rightarrow -2 < y < 2 \Rightarrow -2 < -1 < 2$$

$$\text{Case 2} \rightarrow -2 < y < 1 \Rightarrow -2 < -1 < 1$$

$$\text{Case 3} \rightarrow -3 < y < 1 \Rightarrow -3 < -1 < 1$$

$$\text{Case 4} \rightarrow -4 < y < 1 \Rightarrow -4 < -1 < 1$$

Here, case 2 is more accurate.

- Q.59** A student attempted to solve a quadratic equation in x twice. However, in the first attempt, he incorrectly wrote the constant term and ended up with the roots as (4, 3). In the second attempt, he incorrectly wrote down the coefficient of x and got the roots as (3, 2). Based on the above information, the roots of the correct quadratic equation are
- (A) (-3, 4) (B) (3, -4) (C) (6, 1) (D) (4, 2)

Answer (C)

Solution .

In first attempt:-

$$\alpha + \beta = \frac{-b}{a}$$

And roots (4,3)

$$\therefore 4 + 3 = \frac{-b}{a}$$

$$7 = \frac{-b}{a}$$

$$\text{Or } b = -7a$$

In second attempt:-

$$\alpha \cdot \beta = \frac{c}{a}$$

And roots (3,2)

$$\therefore 3 \cdot 2 = \frac{c}{a}$$

$$\text{Or } c = 6a$$

Now,

$$ax^2 + bx + c = 0$$

$$ax^2 - 7ax + 6a = 0$$

$$x^2 - 7x + 6 = 0$$

$$x^2 - 6x - x + 6 = 0$$

$$\therefore x = 1, 6$$

Q.60 L, M and N are waiting in a queue meant for children to enter the zoo. There are 5 children between L and M, and 8 children between M and N. If there are 3 children ahead of N and 21 children behind L, then what is the minimum number of children in the queue?

(A) 28

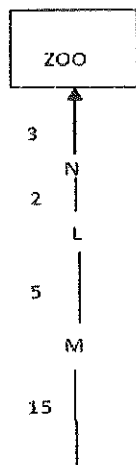
(B) 27

(C) 41

(D) 40

Answer (A)

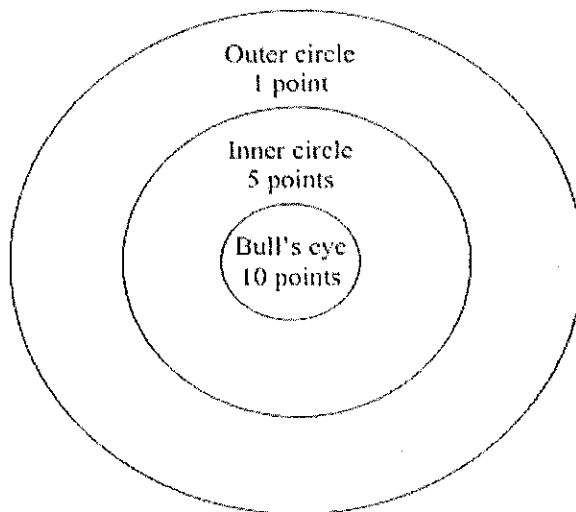
Solution .



∴ minimum number of children in the queue :-

$$3 + N + 2 + L + 5 + M + 15 = 28$$

- Q.61** Four archers P, Q, R and S try to hit a bull's eye during a tournament consisting of seven rounds. As illustrated in the figure below, a player receives 10 points for hitting the bulls' eye, 5 points for hitting within the inner circle and 1 point for hitting within the outer circle.



The final scores received by the players during the tournament are listed in the table below.

Round	P	Q	R	S
1	1	5	1	10
2	5	10	10	1
3	1	1	1	5
4	10	10	1	1
5	1	5	5	10
6	10	5	1	1
7	5	10	1	1

The most accurate and the most consistent players during the tournament are respectively

(A) P and S

(B) Q and R

(C) Q and Q

(D) R and Q

Answer (B)

Solution .

	P	Q	R	S
$E(X)$	4.72	6.57	2.86	4.14
$[E(X)]^2$	22.29	43.17	8.18	17.14
$E(X^2)$	36.14	53.72	18.57	32.72
$\text{Var}(X)$	13.85	10.55	10.39	15.58

Most accurate = max. $E(X)$ value

= Q is most accurate

Most consistent = min. $\text{Var}(X)$

= R is most consistent

Q.62 Nimbus clouds are dark and ragged, stratus clouds appear dull in colour and cover the entire sky. Cirrus clouds are thin and delicate, whereas cumulus clouds look like cotton balls.

It can be inferred from the passage that

- (A) A cumulus cloud on the ground is called fog**
- (B) It is easy to predict the weather by studying clouds**
- (C) Clouds are generally of very different shapes, sizes and mass**
- (D) There are four basic cloud types: stratus, nimbus, cumulus and cirrus**

Answer (D)